

# Digital Platform for Social Welfare Services Management in Sri Lanka

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# Our Team

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# Research Problem

Sri Lanka's welfare services rely on manual processes, requiring citizens to visit service centers, submit documents, and wait in long queues. This results in delays, errors in data collection, limited government staff, and poor accessibility, particularly for elderly individuals and residents in remote areas, reducing efficiency and service quality.



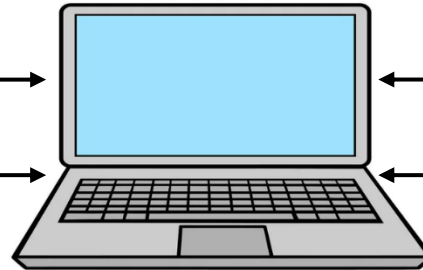
# Research Objectives

## Investment Plan Recommendation

Recommend financial support or investment options to users (e.g., microloans or savings plans).

## Eligibility Prediction & Healthcare Recommendation

Predict who qualifies for welfare programs and suggest hospitals or doctors (e.g., for pregnant mothers or elderly patients).



## AI-GIS Welfare Insights Platform

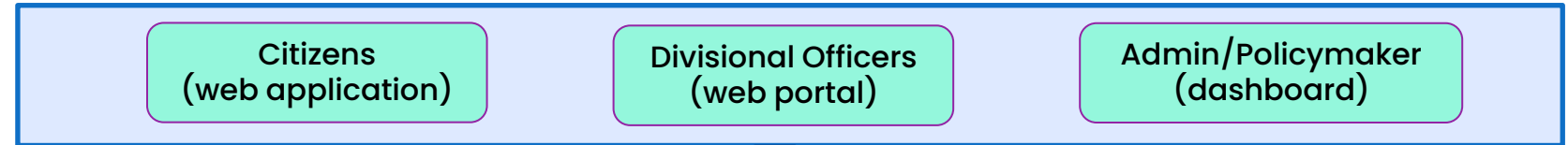
Map areas with high welfare demand to help policymakers (e.g., regions with many chronic patients or low-income families).

## Multilingual Conversational Assistance

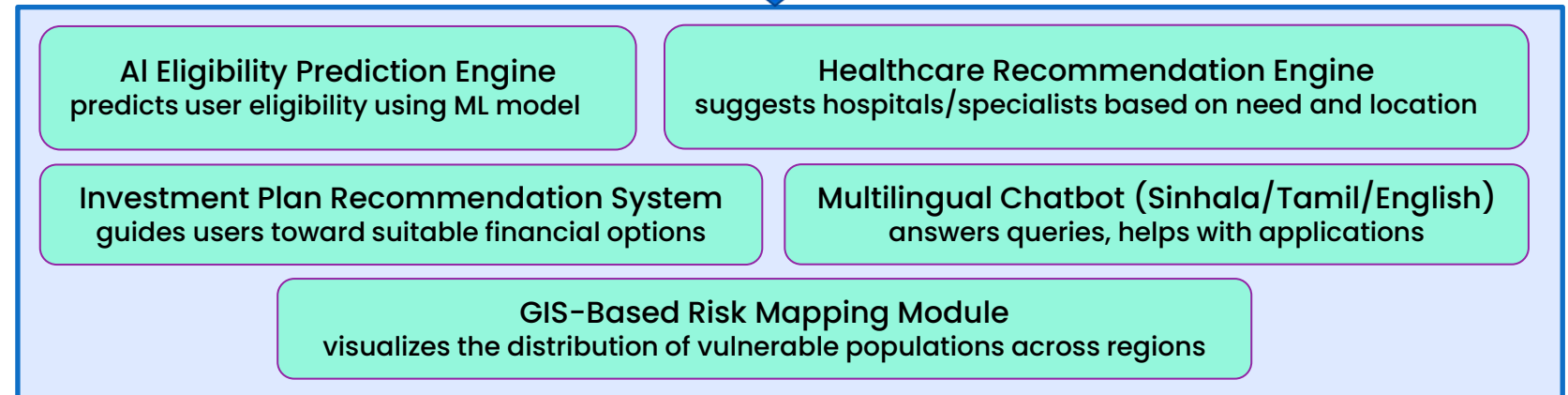
Build a chatbot in Sinhala, Tamil, and English to guide users on applications and documents (e.g., check application status or required forms).

# Overall System Diagram

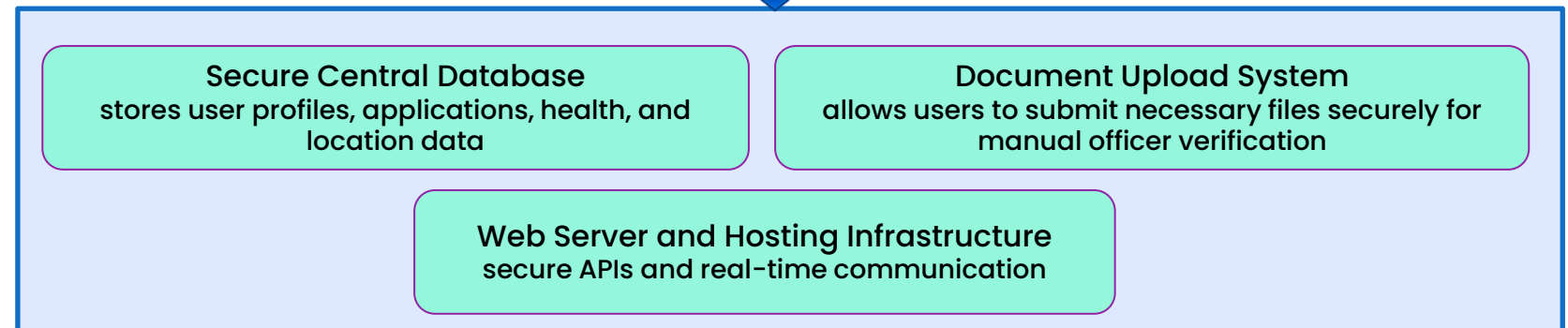
User Interface Layer



AI Intelligence Layer



Infrastructure Layer



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## **Ai-based Investment Plan Recommendation Module For Digital Welfare Management In Sri Lanka**

# Research Problem

How can AI-driven clustering and recommender systems help citizens access suitable investments and financial aid schemes, improving financial decision-making and long-term stability?

Lack of personalized financial advice for users

No AI-based recommendations for investments and aid

Limited focus on user risk and income patterns



# Research Gap

| Aspect                      | Existing Systems                         | Gap                                                     | Our Solution                                    |
|-----------------------------|------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| Financial Guidance          | Generic banking or NGO advisory programs | Not tailored to low-income citizens                     | AI-driven personalized financial advice         |
| Risk & Income Consideration | Basic income checks only                 | Limited analysis of risk profiles and spending patterns | AI models analyzing income, expenses, and risk  |
| Accessibility & Scale       | Manual counseling, time-consuming        | Not scalable for large citizen groups                   | Automated AI module offering wide accessibility |



# Previous Research And Shortcomings

## Previous Research

Research on financial inclusion focused on improving literacy and expanding banking access.

**FinTech** research applied AI to fraud detection, credit scoring, and market predictions.

## Shortcomings

Aid platforms give generic advice, ignoring income, expenses, or risks.

FinTech apps often overlook AI support for low-income investment planning.

# Methodology

## 1. Data Collection

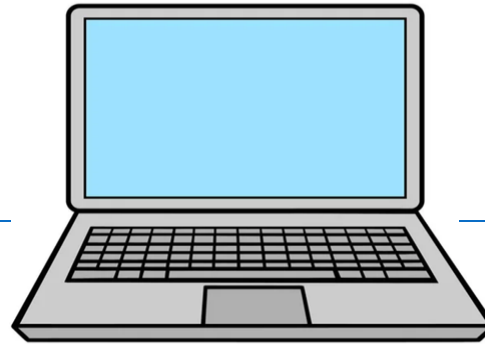
Gather anonymized user income, expenses, and available financial aid schemes.

## 2. Data Preprocessing

Clean and prepare data for analysis, like handling missing values and normalizing numbers.

## 3. User Profiling & Clustering

Group users with similar financial behaviors using clustering techniques.



## 4. Recommendation System

Suggest suitable micro-investments or aid schemes based on user cluster and preferences.

## 5. Validation & Testing

Check if recommendations are accurate and safe for users.

## 6. Deployment

Make the module usable so users can input data and get recommendations.

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## Design and Development of a Multilingual Chatbot for Enhancing Accessibility

# How the Solution Addresses the Sub-Problem

## Sub-Problem

- No quick support in Sinhala and Tamil.
- Long waiting time for call centers.
- FAQ details are in many places.

## Solution

- AI chatbot gives **24/7 quick replies.**
- Supports **Sinhala, Tamil, and English.**
- All FAQs and help details in **one place.**

## Result

- Less load on call centers.
- Better customer satisfaction.

## Prototype User Requirements

- Ask questions in Sinhala, Tamil, or English.
- Quick answers for data, account, and network issues.
- Easy-to-use chat interface.
- Accurate and consistent responses.

**All implemented in the prototype chatbot**



# Solution Design Excellence / Contribution

## Design

- Multilingual NLP with language detection
- Scalable, knowledge-base driven (PDF/JSON)

## Contribution

- Boosts digital inclusiveness
- Supports local languages
- Reduces service costs



## User Feedback on Prototype

- Communicate in native language
- Faster than call centers
- Simple, user-friendly interface
- More detailed service info





# Commercialization / Sustainability of Solution

- Can be sold as a Software-as-a-Service (SaaS) solution.
- Can be customized for:
  - ✓ Telecom companies
  - ✓ Banks
  - ✓ Government services
- Low maintenance cost after deployment.
- Continuous improvement using user interaction data.
- Ensures long-term sustainability and scalability.



## Customer Persona

### ❑ Primary Customers:

- ✓ Telecom service providers (e.g., Dialog, Mobitel, Airtel)
- ✓ Sri Lankan users who prefer Sinhala or Tamil

### ❑ Persona Example:

Name: Kamal / Arjun

Age: 25–55

Uses mobile services daily

Needs quick support without calling customer care



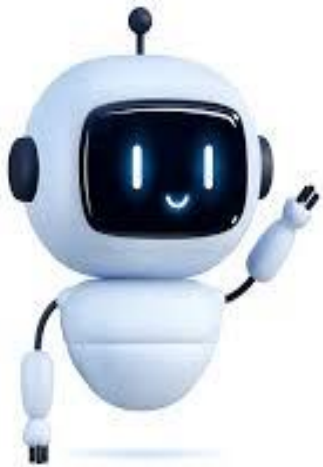
## Market Size

- Over 22 million mobile subscribers in Sri Lanka.
- Millions of daily customer support inquiries.
- Growing demand for AI-based automation.
- High potential in telecom, banking, and public services sectors.

**Large and growing market opportunity.**

## What is Unique in Your Solution

- Strong focus on **Sinhala and Tamil language support.**
- Tailored for Sri Lankan context.
- Lightweight and cost-effective solution.
- Easily expandable to other domains.
- Can operate even with limited resources



## How Will the Cost Be Recovered?

- Subscription-based pricing for organizations.
- One-time setup and customization fees.
- Maintenance and support contracts.
- Licensing the chatbot to multiple companies.

**Ensures revenue generation and cost recovery.**

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## AI-GIS Integrated Vulnerable Population Monitoring and Aid Recommendation System for Welfare Services

# Research Problem

How can an AI–GIS integrated system be designed to accurately map vulnerable populations and provide data-driven welfare aid recommendations in Sri Lanka, ensuring fairness, efficiency, and transparency in social service delivery?

Which communities in Colombo District are most in need of welfare support?

How can socio-economic and hazard data be combined into a Social Welfare Need Index (SWNI)?

Can AI and optimization techniques improve the targeting and distribution of welfare aid?



# Research Gap

Welfare identification in Sri Lanka is manual, fragmented, and paper-based..

Current systems lack GIS integration → no spatial prioritization of vulnerable areas.

Data silos between census, social services, and disaster management → inefficient planning.

No predictive analytics for anticipating needs during disasters or economic shocks.

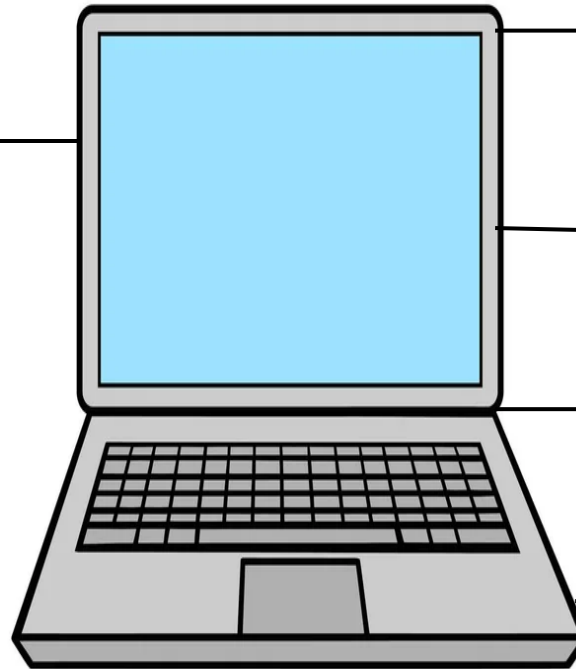
Aid allocation lacks fairness checks, leading to under-served communities.



# Specific and Sub Objectives

## Specific Objectives

To design and implement an AI-GIS integrated system that identifies vulnerable populations and recommends optimized welfare aid distribution.



## Sub Objectives

Develop a geospatial database combining welfare, demographic, and hazard data.

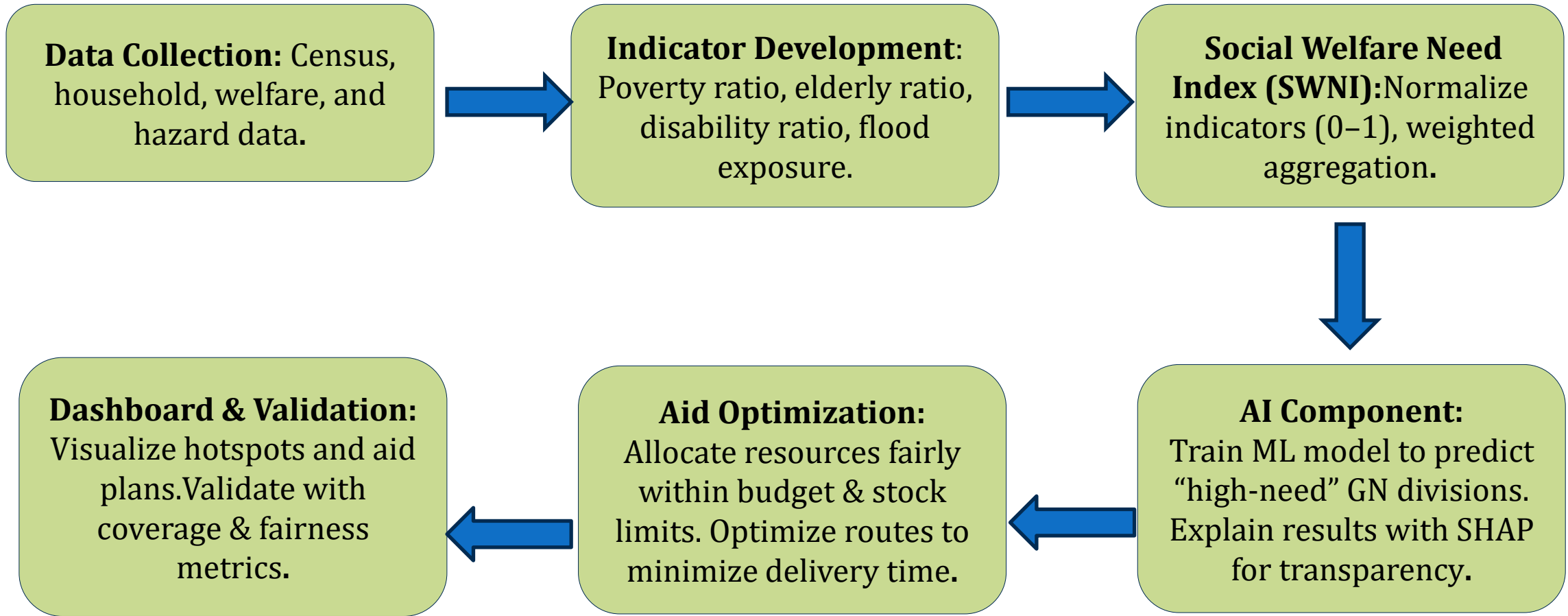
Build a Social Welfare Need Index (SWNI) to rank vulnerability at GN/DS level.

Integrate AI models to predict welfare needs and prioritize groups.

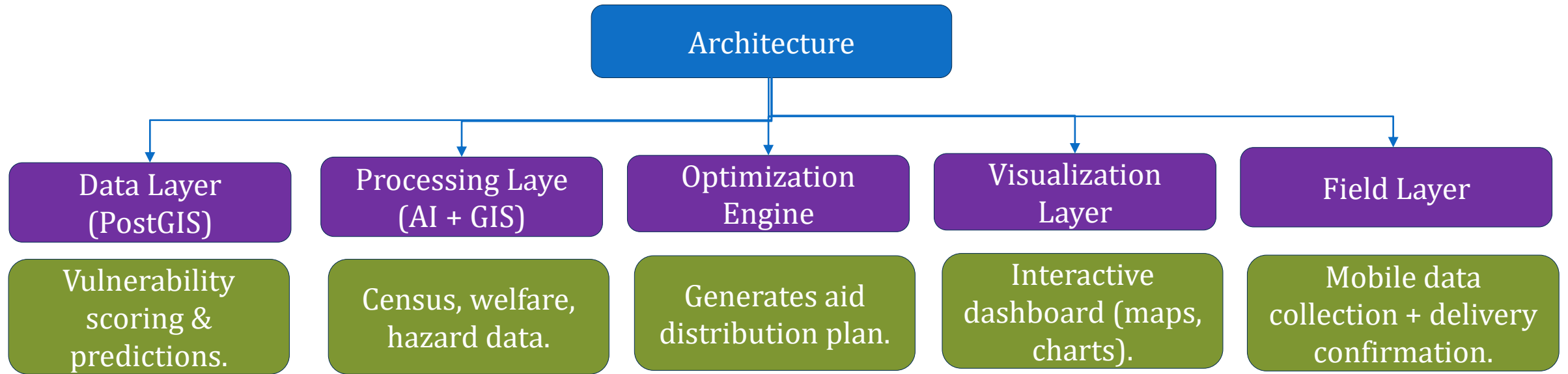
Optimize aid allocation & routing considering budget, stock, and accessibility.



# Methodology



# System Overview



## Output

GN-level SWNI maps.

Priority ranking of communities..

Optimized aid plans with routes.



# Real World Use Case

## Flood in Kolonnawa DS Division (Colombo)

System highlights GN divisions at highest risk (e.g., Sedawatta, Meethotamulla).

Welfare & hazard data combined → High SWNI zones flagged red.

AI model predicts extra households needing support.

Optimizer generates aid distribution plan:

200 dry-ration packs → 3 GN divisions.

2 routes optimized from main warehouse (time reduced by 30%).

Field officers receive assignment on mobile app → confirm delivery → update system .

Impact: Faster, fairer, data-driven welfare response.



# Completion of the project

Data Integration: Combined census, welfare, and hazard datasets into a PostGIS spatial database with privacy safeguards

Vulnerability Mapping: Built a Social Welfare Need Index (SWNI) and produced GN-level heatmaps..

Outcomes: A scalable, ethical AI-GIS platform improving fairness, efficiency, and transparency in welfare services..



AI Component: Applied ML models (XGBoost, Logistic Regression) + Explainable AI (SHAP) for predicting vulnerable areas

Aid Optimization: Developed an allocation & routing engine (OR-Tools, OSRM) for fair and efficient distribution.

System Prototype: Built a dashboard (FastAPI + React/Leaflet) with maps, rankings, and aid plans

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## AI-Driven Eligibility Prediction and Healthcare Recommendation System

# How the Solution Addresses the Sub-Problem

## The Problem

- Current welfare targeting in Sri Lanka often relies on static, manual criteria that lead to misallocation

## The Solution

- Developed a real-time Citizen Eligibility & Healthcare Recommendation System using Machine Learning.

## Result

- The **Random Forest model** achieved **100% accuracy** on the test dataset and was selected as the core engine.

## Methodology

- Evaluated four architectures (Logistic Regression, SVM, KNN, and Random Forest) to ensure scientific accuracy.

## User Requirements Addressed

Automated Eligibility: Instantly determines "Eligible" or "Not Eligible" status based on socio-economic inputs like income and family size.

Smart Medical Referral: Automatically recommends public vs. private facilities based on the citizen's predicted welfare status.



# Solution Design Excellence / Contribution

## Design Excellence / Contribution

- Completed Front End:

Developed a fully functional portal that allows live interaction with the ML model.

- Integration Ready:

Built with a modular architecture to receive data from the group's Chatbot and send results to the GIS module.

- Logic Innovation:

Implemented "Financial Priority" logic where welfare-eligible citizens are automatically prioritized for free Government Teaching Hospitals.

## User Feedback on Prototype

- Speed:

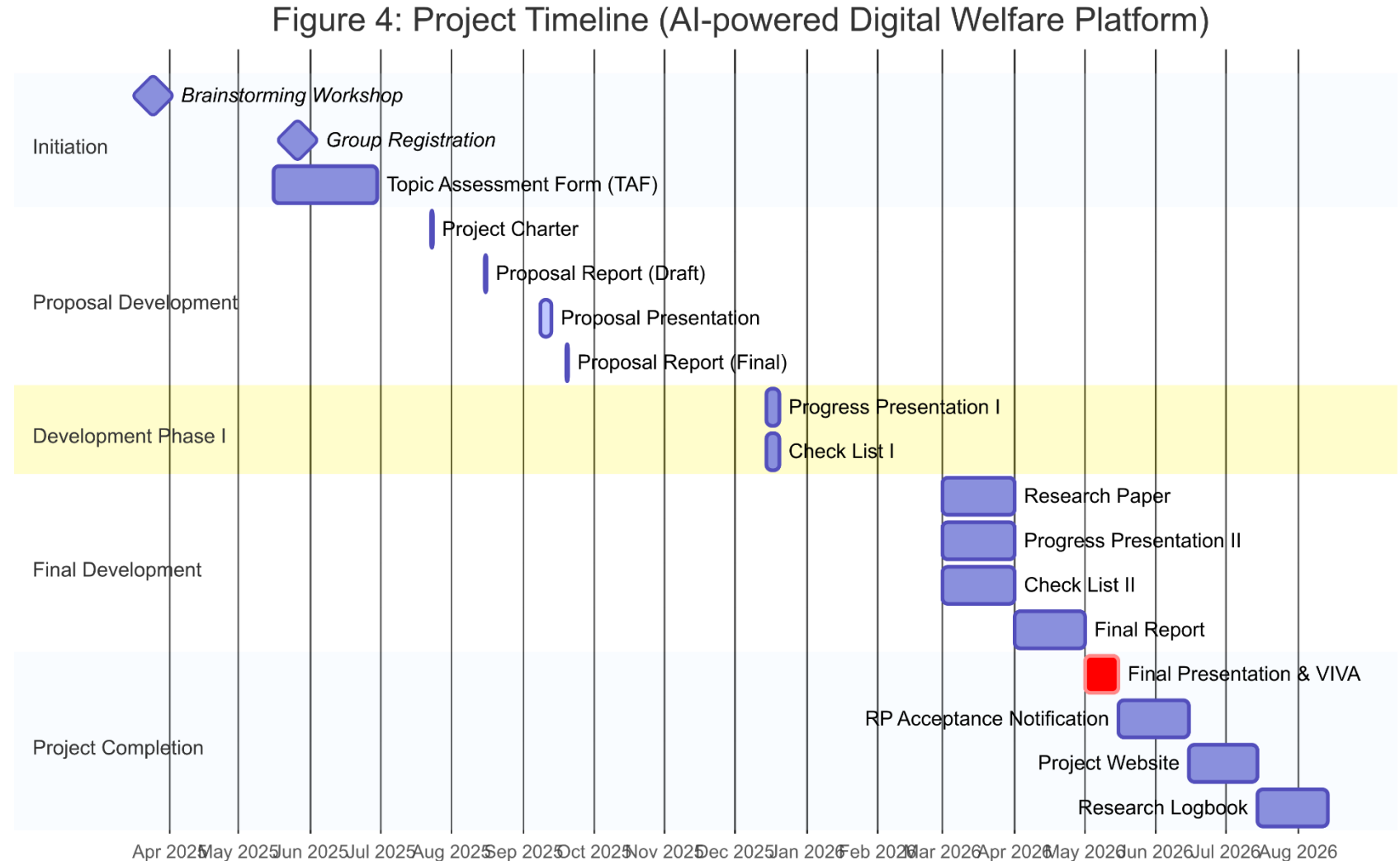
Users noted that instant eligibility feedback reduces the "waiting stress" associated with traditional paper-based systems.

- Clarity:

The visual layout (Results & Medical Facilities columns) makes complex welfare rules easy to understand for the average citizen.

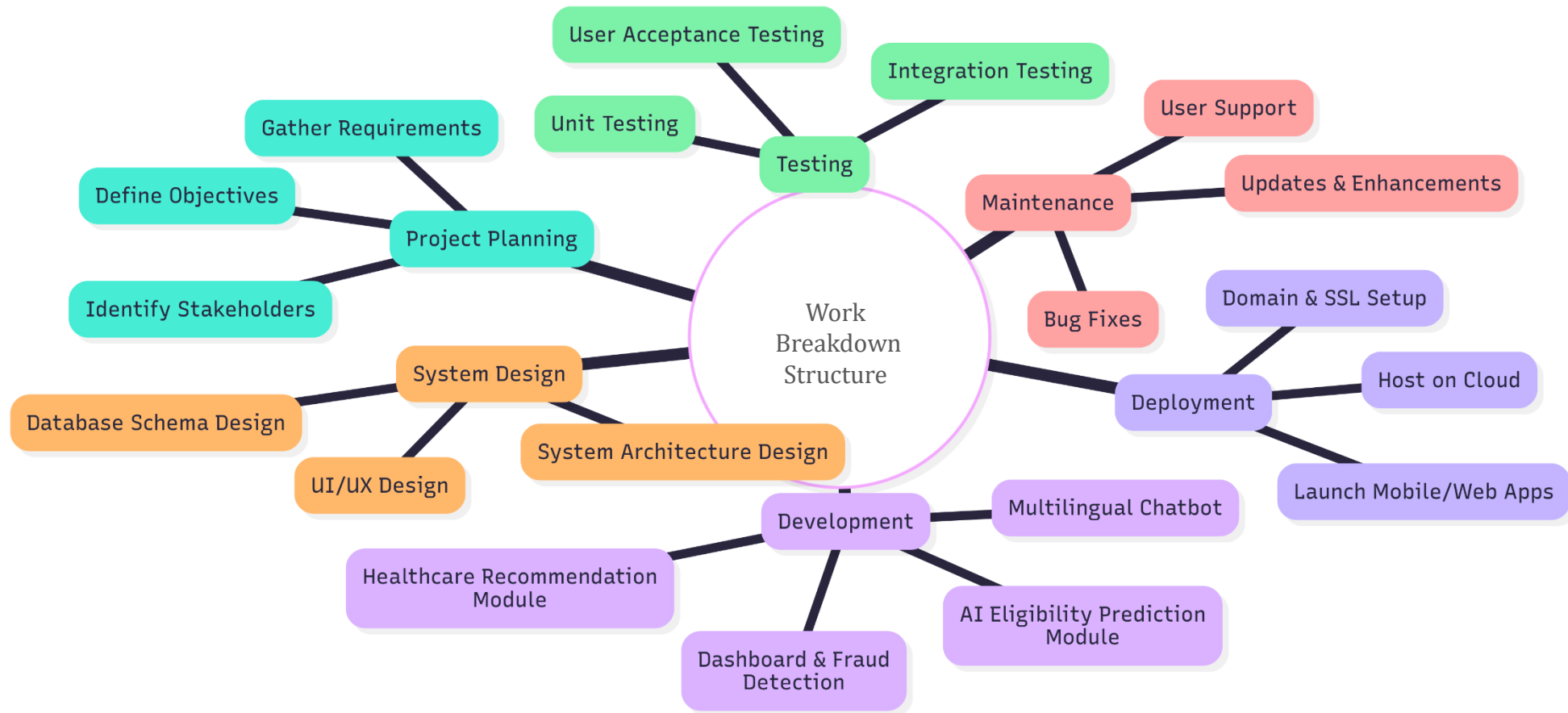
# Completion of the project

## Gantt Chart



# Completion of the project

## Work Breakdown Structure



# References

- [1] M. U. Ahmed, AI Applications in Social Assistance Programs, South Asian Journal of Policy and Governance, 2019.
- [2] S. Bhatnagar, E-Government: From Vision to Implementation. SAGE Publications, 2004.
- [3] World Bank. (2021). Sri Lanka Poverty Assessment. Washington, DC: World Bank.
- [4] Demirguc-Kunt, A., Klapper, L., Singer, D., & Ansar, S. (2018). The Global Findex Database 2017: Measuring Financial Inclusion and the Fintech Revolution. World Bank.
- [5] M. U. Ahmed, "Digital transformation of social welfare services: Challenges and opportunities in South Asia," South Asian Journal of Policy and Governance, 2019.
- [6] United Nations, E-Government Survey 2022: Digital government in the decade of action for sustainable development. New York, NY, USA: United Nations, 2022.
- [7] M. U. Ahmed, "AI Applications in Social Assistance Programs," South Asian Journal of Policy and Governance, 2019.
- [8] S. Bhatnagar, E-Government: From Vision to Implementation. SAGE Publications, 2004.
- [9] United Nations, E-Government Survey 2022: The Future of Digital Government. New York, NY, USA: United Nations, 2022.

